Low-Level Design (LLD)

RAG-Based Customer Support Assistant

1. Module Design

Document Processing Module

Loads PDF and extracts text.

Chunking Module

Splits text into chunks with overlap.

Embedding Module

Generates embeddings using sentence transformers.

Vector Storage Module

Stores embeddings in ChromaDB.

Retrieval Module

Fetches top-K relevant chunks.

Query Processing Module

Generates answers using LLM.

Graph Execution Module

Executes workflow using LangGraph.

HITL Module

Handles escalation and human response.

2. Data Structures

Document

{
"doc_id": "1",
"text": "content"
}

Chunk

{
"chunk_id": "c1",
"text": "...",
"metadata": { "page": 1 }
}

Graph State

{
"query": "",
"docs": [],
"answer": "",
"confidence": 0.0,
"escalate": false
}

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3. Workflow Design

Nodes:

- Processing Node
- Output Node

Flow:
Input → Processing → Output

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4. Conditional Routing

If:

- confidence < threshold → escalate

no docs → escalate

else → answer

5. HITL Design

Trigger:

Low confidence

Missing context

Flow:
AI → Human → Response

6. API Design

Input:
{
"query": "question"
}

Output:
{
"answer": "...",
"confidence": 0.85,
"source": "AI/Human"
}

7. Error Handling

No data → escalate

LLM failure → retry

Invalid input → validation

Conclusion

The LLD defines internal implementation details ensuring modularity and scalability.